

0.1 Problem Setup

Let $D(t)$, $S(t)$, $E(t)$ be water demand, solar supply, and electric heating rate (volume per unit time). Tank capacity is X .

Constraint (for all intervals T):

$$\int_T (D(t) - S(t) - E(t)) dt \leq X$$

Objective: minimize total electric heating

$$\int E(t) dt$$

0.2 Robust Receding-Horizon Controller

Discretize time with step Δt . At each time k :

- Current cumulative deficit: $U_k = \int_0^{t_k} (D - S - E) dt$ - Historical minimum:

$m_k = \min_{s \leq t_k} U(s)$ - Remaining buffer: $r_k = X - (U_k - m_k)$

Forecast horizon: H steps (≈ 1 hour).

Use worst-case forecasts over uncertainty sets:

$$\hat{D}_{k+i}^w = \hat{D}_{k+i} + \delta_D(i), \quad \hat{S}_{k+i}^w = \hat{S}_{k+i} - \delta_S(i)$$

Solve the following linear program at each k :

$$\begin{aligned} & \text{minimize} && \sum_{i=1}^H E_i \Delta t \\ & \text{subject to} && E_i \geq 0, \quad i = 1, \dots, H \\ & && U_{k+i} = U_k + \sum_{j=1}^i (\hat{D}_{k+j}^w - \hat{S}_{k+j}^w - E_j) \Delta t \\ & && m_0 = m_k \\ & && m_i \leq m_{i-1}, \quad m_i \leq U_{k+i} \quad \forall i = 1, \dots, H \\ & && U_{k+i} \leq m_i + X \quad \forall i = 1, \dots, H \end{aligned}$$

Apply only the first decision: $E(t) = E_1^*$ for the next Δt . Update U and m with actual measurements at the next step and re-solve.

This ensures the tank constraint holds for all possible realizations within the uncertainty bounds while minimizing electric heating.